

CLIENT WORKFLOW AND FULL-STACK REVIEW

Marble Park Client Workflow Readiness

EXECUTIVE SUMMARY

The client requests are valid and should be handled as one commercial workflow. Product editing, quote pricing, and partial order conversion share the same product and line snapshots; implementing them independently would create data drift.

Direct quote creation already works. Quote Studio accepts a customer and product lines without a lead or intent, while the API creates the internal CRM opportunity automatically.

Product editing, editable quote rate, and partial sales orders are not end-to-end complete. Product updates exist only in the API, quote-rate totals disagree between surfaces, and the database enforces one sales order and one dispatch job per quote.

Production is currently offline. Railway reports both application services stopped with no running replicas, so restoring a controlled deployment is the first operational action before client validation.

Direct quote

Existing behavior; needs a dedicated regression test.

Product edit

API capable; no usable Product Master edit screen.

Quote rate

Visible calculation and stored order value can disagree.

Partial order

Blocked by one-order-per-quote data constraints.

Production

Web and API services are stopped.

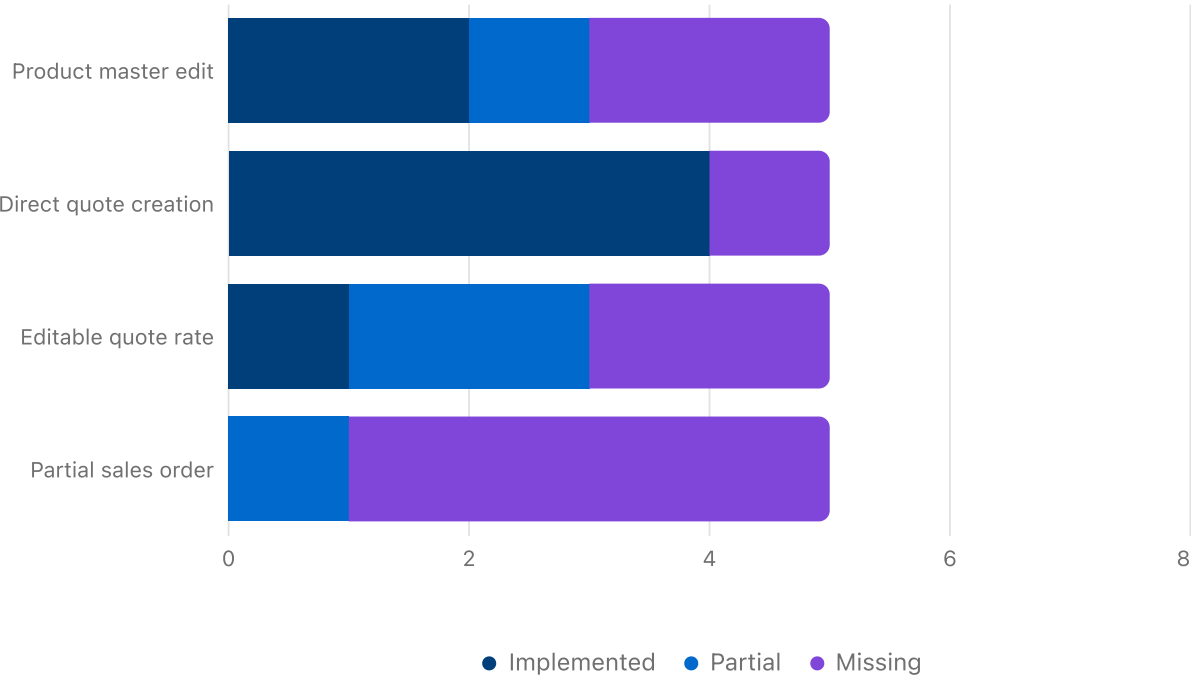
Only direct quote creation is functionally ready

The chart grades each request across UI, API, data model, downstream integrity, and automated test coverage. It is a coverage count, not a subjective percentage. Product editing has backend foundations, pricing has partial UI and schema support but unsafe downstream totals, and partial ordering needs coordinated changes across the full chain.

Caveats and Assumptions. This is a no-change audit of the current main branch and read-only Railway state on July 10, 2026. Local production builds and Prisma validation passed. Mutating smoke scripts were not run, and production was offline, so authenticated live workflows could not be executed. No application code, migration, database record, or hosting configuration was changed.

Requirement coverage by delivery layer

Five audited layers per request: UI, API, data model, downstream integrity, and automated test.



Coverage is based on traced repository behavior, current schema constraints, available smoke scripts, and read-only Railway status.
Evidence: client message; Product Master, quote, order, stock, and dispatch source; Prisma schema; smoke scripts.

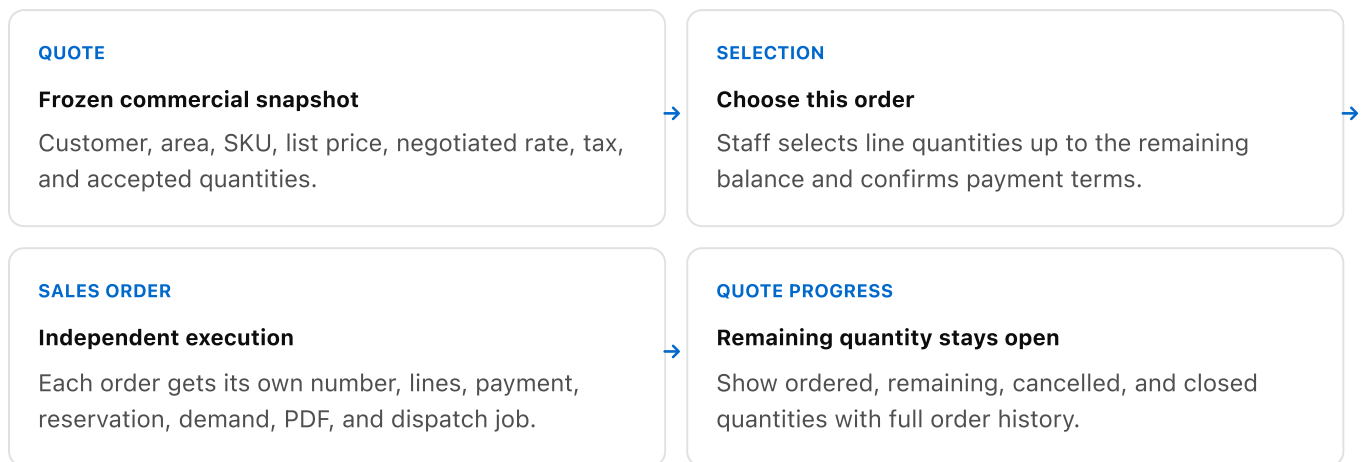
Client request to current-system mapping

Current behavior and the minimum complete outcome for client acceptance.

Client request	Status	Current system	Complete acceptance outcome
Product master editable; SKU fixed	Partial	API update excludes SKU, but Product Master has create-only UI; archive can still rewrite SKU.	Edit all commercial fields and image gallery; SKU read-only; diff audit; archive without changing code.
Quote without lead or intent	Ready	Quote Studio creates from customer and lines; API auto-creates the internal opportunity.	Keep this path, label it clearly, and add a regression test for no-lead creation.
Rate available while quoting	Unsafe	Creation shows catalogue rate read-only; detail edits MRP and discount; order totals ignore special-rate semantics.	Editable line rate at creation and detail, one canonical pricing function, floor-price policy, identical UI/PDF/order totals.
Multiple partial orders from one quote	Missing	Partial challans exist, but quote-to-order conversion always uses all lines and quote ID is unique on sales order and dispatch job.	Select line quantities, create repeated orders against remaining quantity, and isolate stock, payment, procurement, documents, and dispatch per order.

The correct design keeps one quote and allows many controlled orders

The quote remains the negotiated customer record. Each confirmed subset becomes its own sales order, with independent payment, reservation, procurement, document, and dispatch state. Remaining quote quantities stay available for the next order instead of forcing a duplicate quote.



Concurrency rule: sales-order creation must be transactional and idempotent. Two staff members cannot order the same remaining quantity, and retrying a request cannot create a duplicate order.

Pricing must become one business rule, not three calculations

The current quote UI and PDFs understand special rate and line discount, while sales-order totals and normalized line prices use catalogue price. The fix should introduce one shared pricing contract used by the browser, API, quote PDF, sales-order PDF, payments, and reports.

- **Line snapshot:** list price, discount percentage, final unit rate, tax rate, taxable value, tax amount, and gross line total.
- **Commercial guardrail:** below-floor rate requires an explicit permission or owner approval, with the requested and approved rate recorded.
- **Version safety:** editing Product Master changes future quotes only; existing quote versions and sales orders keep their historical rate and image snapshot.
- **Single calculator:** persist server-calculated totals and make both PDFs read those values instead of recalculating independently.

Product Master needs a real edit lifecycle

The backend already has the right first boundary because the update input does not accept SKU. The complete workflow should add an edit action from the SKU register and product detail, expose the existing commercial fields, and close audit and image-management gaps.

- **Immutable identity:** SKU is read-only in UI and rejected if supplied to update APIs; archive keeps the original SKU.
- **Editable content:** name, category, brand, finish, dimensions, unit, sell price, floor price, tax class, description, status, primary image, and gallery.
- **Image controls:** upload, replace, remove, set primary, reorder gallery, file validation, size limits, and durable storage.
- **Audit and conflict control:** record field-level before/after values and reject stale edits using a version or updated timestamp.

Full-stack review found four release blockers beyond the UI

CRITICAL Production hosting

Railway shows both application services failed/stopped, zero running replicas, and no connected source repository. Public web and API domains return Application not found.

CRITICAL Pricing integrity

Special rate and discounts are calculated in quote UI/PDF, but order totals, normalized unit price, and payment status use catalogue price. This can create customer-document and accounting disagreement.

CRITICAL Order data model

Sales order and dispatch job both enforce quote ID uniqueness. Reservations and several downstream lookups are quote-scoped, so partial ordering cannot be made safe with a frontend-only change.

CRITICAL Security and uploads

Production GraphQL playground is enabled, sessions live in browser storage, login has no throttling, broad roles can upload arbitrary public files, and request bodies permit very large payloads.

HIGH Dependency risk

The current production dependency audit reports six high and four moderate advisories, including websocket, upload, mail, and CSS processing paths.

HIGH Automated verification

Production builds and schema validation pass, but there are no unit or integration test files. Existing smoke scripts mutate business data and do not cover product edit, direct quote without lead, canonical rate parity, or multiple orders per quote.

HIGH Database integrity

Several normalized operational tables have IDs and indexes but no foreign keys in migrations. Partial-order work should add explicit relations and deletion rules instead of increasing soft references.

HIGH Delivery discipline

Service source is disconnected, health checks are absent, Docker builds use install rather than lockfile-clean install, and documentation describes older framework and authentication choices.

User database review: 37 of 98 accounts are active in the local snapshot

A read-only aggregate query was run against the configured local `marble_park` database. No names, email addresses, phone numbers, password hashes, tokens, or permission details were extracted.

Account status

98 total users: 37 active and 61 inactive. The recorded account creation range is April 29 to June 19, 2026.

Active role mix

27 office staff, 4 sales, 2 owners, and one each for admin, dispatch operations, inventory manager, and sales manager.

Inactive concentration

All 61 inactive accounts are concentrated in three roles: 31 sales, 15 owners, and 15 dispatch operations. These should be reconciled against current staff before production restore.

Authentication hygiene

All 751 stored sessions are expired and none are active. Password-change timestamps exist for 82 users, while 16 are missing; three users have permission overrides.

Production user counts remain unverified. Railway variables show a TCP proxy correctly targeting PostgreSQL port 5432, but the Postgres service has no active deployment and both public-proxy and SSH-tunnel connections closed before authentication. Restarting or redeploying production would be a state change, so it was not performed in this audit.

End-to-end acceptance must prove business truth across every surface

A release is complete only when UI, API, database state, documents, stock, payments, procurement, dispatch, and audit agree. The acceptance suite should use an isolated test database and disposable uploaded assets, never the client dataset.

Required acceptance scenarios

Minimum automated coverage before client handoff.

Workflow	Cases that must pass	Pass condition
Product edit	Edit every commercial field, replace and reorder images, attempt SKU change, stale-edit conflict, permission denial, archive.	SKU never changes; catalogue and future quote search refresh; audit contains exact diff.
Direct quote	Create from customer with no lead or intent; assign owner; reopen and revise.	Quote saves, internal CRM link exists, PDF renders, and no user-facing lead step is required.
Rate parity	Catalogue rate, negotiated rate, line discount, quote discount, tax, floor-price breach, revision.	UI, API, quote PDF, order PDF, payment balance, and reports return the same totals.
Partial orders	Order a subset, order remaining lines later, partial quantity, concurrent over-order attempt, retry, close remainder.	Ordered plus remaining plus cancelled equals quoted; no duplicate or over-order is possible.
Downstream isolation	Separate cash/credit payments, reservation, backorder demand, GRN allocation, dispatch, delivery, return, and cancellation for each order.	Every operational record points to the correct sales-order line and stock reconciliation stays clean.
Deployment	Migration rehearsal, rollback rehearsal, dependency gate, health checks, authenticated browser journey, document downloads.	Connected deployment becomes healthy, source revision is traceable, and smoke evidence is saved without touching client data.

Adjacent features the client is likely to expect

- **Quote fulfillment view:** ordered, remaining, cancelled, reserved, pending inward, dispatched, and delivered by line and area.
- **Price and product history:** effective date, prior value, changed by, reason, and future-price scheduling.
- **Duplicate active-quote warning:** alert staff when the same customer and project already has an open quote; open the current quote instead of creating another.
- **Close or cancel remainder:** release stock and purchase demand with a reason, preserving the accepted order history.
- **Per-order promise and payment:** delivery date, payment terms, advance allocation, outstanding amount, and credit approval on each partial order.
- **Inline customer creation and quote draft recovery:** create the customer from Quote Studio, autosave drafts, and recover interrupted work.

Recommended Next Steps

1. **Restore a controlled production deployment first.** Reconnect Railway services to source, configure health checks, deploy the current main revision, and verify login plus read-only core queries before client testing.
2. **Fix canonical pricing before exposing editable rate.** Introduce the server pricing contract and parity tests, then make Rate editable in both Quote Studio and quote detail.
3. **Complete Product Master editing.** Add immutable-SKU edit UI, audit history, archive behavior, safe image management, and focused tests.
4. **Migrate to one quote with many sales orders.** Change order, reservation, dispatch, payment, procurement, and document ownership together, with backfill and rollback scripts.
5. **Run the isolated end-to-end release gate.** Only after that gate passes should the client dataset be migrated and the client acceptance session begin.

Decisions to confirm with the client

- Recommended default: only admin, owner, and inventory manager can edit Product Master; sales can view and quote.
- Recommended default: a rate below floor price requires owner approval before PDF send or order creation.
- Recommended default: each partial order has its own payment terms, delivery promise, PDF, and dispatch job.
- Recommended default: remaining quote quantity stays open until fully ordered or explicitly closed with a reason.